

Commercial TnH+IAQ Sensor with Modbus RTU/WIFI-MQTT

Communication – Datasheet

1. Product Overview

WIN-SN-TnH-IAQ is a versatile data acquisition device designed for real-time monitoring in industrial applications. Providing flexibility for various measurements. The device integrates seamlessly with **Modbus RTU**, offering reliable communication and long-range data transfer over several meters, making it ideal for large-scale installations.

Its robust design makes it suitable for **installations in HVAC systems, environmental monitoring, and process control**. With **configurable data sampling rates, multi-sensor compatibility, and easy configuration**, WIN-SN-TnH-IAQ ensures **accurate Air-quality, Temperature and Humidity data acquisition and reliable performance** in demanding industrial environments, making it an excellent choice for **industrial and IoT applications**. Additionally, it features a 4-digit 7-segment display that provides real-time measurement data, allowing users to monitor sensor readings directly without requiring an external interface.



Ex - Product Image

2. Technical Parameters

- Measuring Parameters of Air Quality, Temperature, and Humidity.
- Response Time: Less than 2 seconds
- **Modbus RTU** communication

	Sensor Name	Temperature Measuring Range	Humidity Measuring range	AQI/ GAS/ Pressure	Measuring Accuracy
☐	BME280	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa	1.0°C, 3.0 %RH, 1.0 %hPa
☐	BME680	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa & 0 – 500 IAQ	1.0°C, 3.0 %RH, 0.6 %hPa, 15 %IAQ
☐	BME688	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa & 0 – 500 IAQ	0.5°C, 3.0 %RH, 0.6 %hPa, 15 %IAQ
☐	BME690	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa & 0 – 500 IAQ	0.5°C, 3.0 %RH, 0.5 %hPa., 15 %IAQ

	Product Configuration
☐	TnH+IAQ+M
☐	TnH+IAQ+M+D
☐	TnH+IAQ+W
☐	TnH+IAQ+W+D
☐	TnH+IAQ+MRTU+WIFI
☐	TnH+IAQ+MRTU+WIFI+D

3. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*36 mm.
- **Certifications:**



4. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

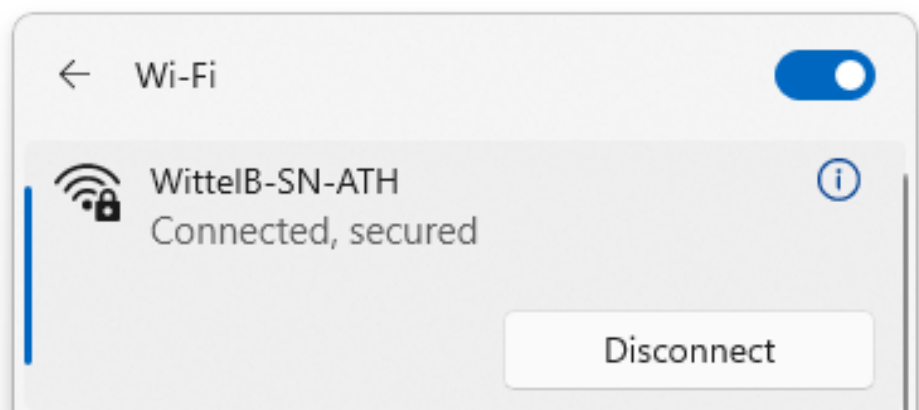
Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

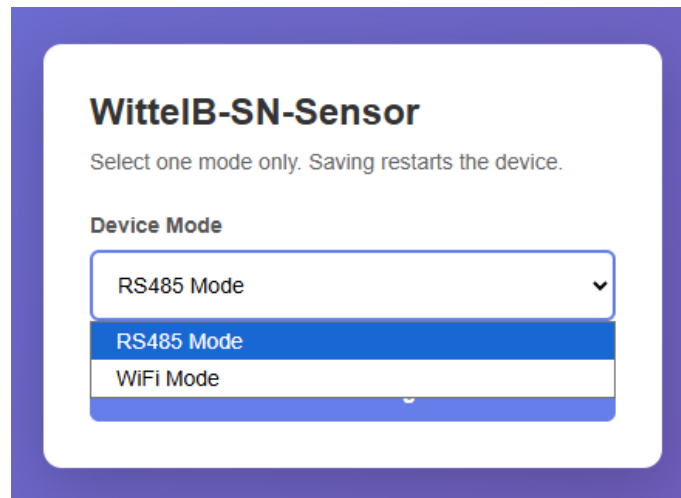
5. Step-by-Step Configuration

1. Connect the Device

- Connect **12v-1A DC** adaptor to power the board
- Connect the **Wi-Fi:-** WittelB-SN-ATH, **Password:-** 123456789
- Search browser enter default **IP : 192.168.4.1**
- The device starts communication using default settings if not previously modified.



2. Mode selection



- Enter **192.168.4.1** in the browser search bar and select the **Device Mode**.
- Select the device mode as **RS485 Mode** to get the sensor data via **Modbus RTU**, or select **Wi-Fi Mode** to get the sensor data in the **MQTT cloud** through the local Wi-Fi network.

1.1 RS485-Mode

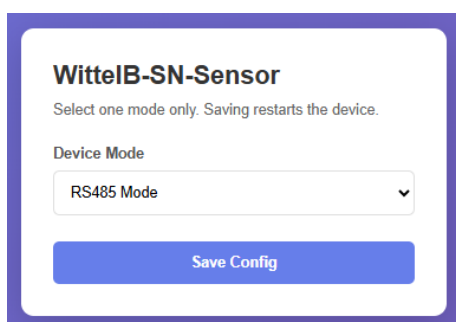


Image: 1

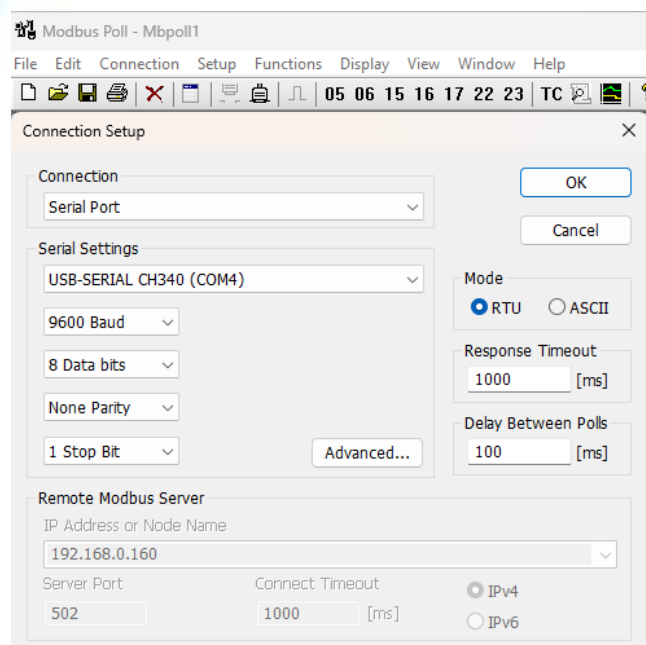


Image: 2

- As shown in **image 1** click **Save Config** and restart the device.
- Refer to **Image 2**. Open **MB Poll/Modbus Scanner** and connect the device using default address.

RS485-Configuration:

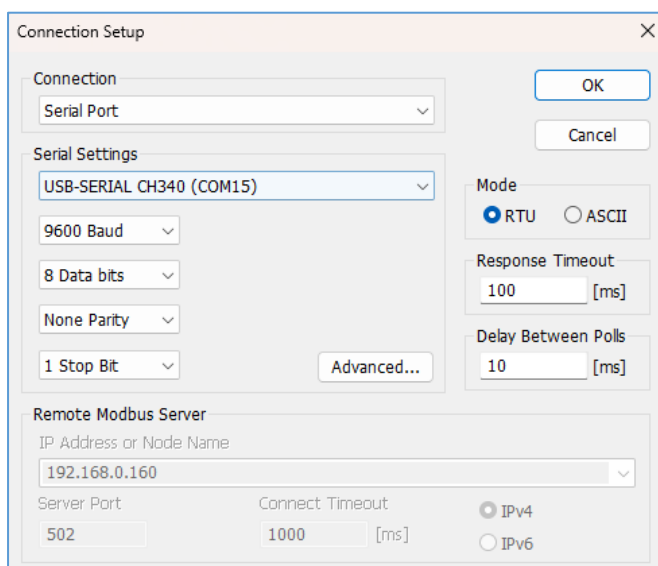
- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None
- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

- Here is an example of receiving data on Mb poll software.
- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).

- After connecting you will get data as shown below.

Important Note

- If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
- After that press the reset button or reboot the system.
- Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.



Connection Setup

Connection: Serial Port [OK] [Cancel]

Serial Settings

USB-SERIAL CH340 (COM15)

9600 Baud

8 Data bits

None Parity

1 Stop Bit

Advanced...

Mode

☒ RTU ☐ ASCII

Response Timeout: 100 [ms]

Delay Between Polls: 10 [ms]

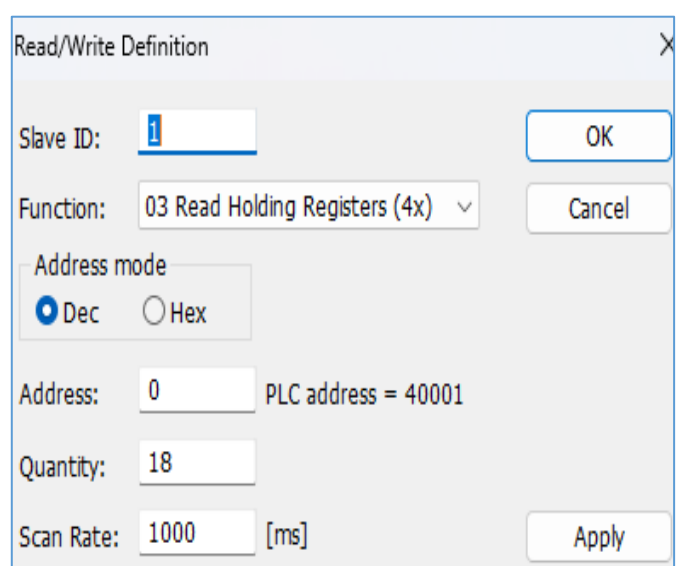
Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502

Connect Timeout: 1000 [ms]

☒ IPv4 ☐ IPv6



Read/Write Definition

Slave ID: 1 [OK] [Cancel]

Function: 03 Read Holding Registers (4x) [OK] [Cancel]

Address mode

☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

Scan Rate: 1000 [ms] [Apply]

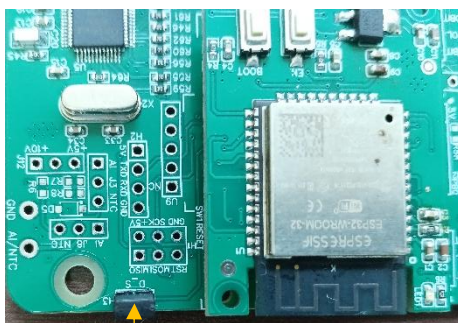
- Above image Shows sensor data over Modbus with default Modbus settings
- Register 40001 Shows the Temperature, Register 40002 Shows the Humidity, Register 40003 Shows the IAQ
- Temperature 326 indicates 32.6, Humidity 458 indicates 45.8
- As you can see in the image above, the resistor 40011 – 960 represents the present baud rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.

Mbpoll1
Tx = 418: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	Temperature	326	Current Baud	960
1	Humidity	458	New Baud	11520
2	IAQ	87	Current SID	1
3		0	New SID	10
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

Mbpoll1
Tx = 418: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	Temperature	326	Current Baud	11520
1	Humidity	458	New Baud	0
2	IAQ	87	Current SID	10
3		0	New SID	0
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0



**RESET
JUMPER**



**Config
JUMPER**

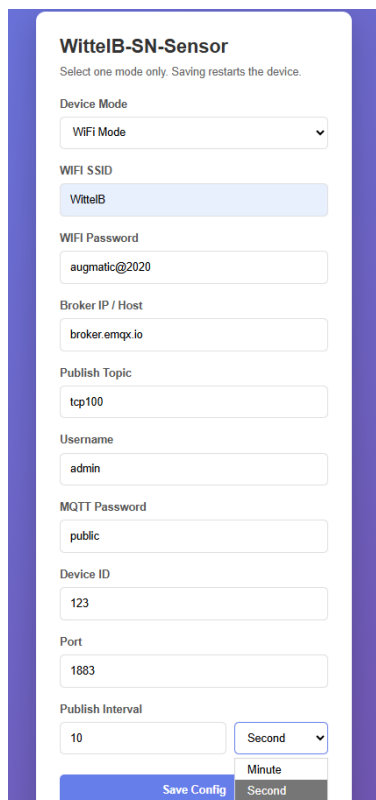
Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the above image 1.1) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

Modbus Registers Mapping

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40001	0x03	RS485	16-bit int
2	Humidity	40002	0x03	RS485	16-bit int
3	IAQ	40003	0x03	RS485	16-bit int
1	Display Baud Rate (Default: 960)	40011	0x03	RS485	16-bit int
2	Enter New Baud Rate	40012	0x03	RS485	16-bit int
3	Display Slave ID (Default: 1)	40013	0x03	RS485	16-bit int
4	Enter New Slave ID	40014	0x03	RS485	16-bit int
5	Display Parity (Default: None-3 And for Odd-1, Even-2)	40015	0x03	RS485	16-bit int
6	Enter New Parity	40016	0x03	RS485	16-bit int
7	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16-bit int
8	Enter New Start/ Stop Bit	40018	0x03	RS485	16-bit int

1.2 WIF-Mode



WittelB-SN-Sensor
Select one mode only. Saving restarts the device.

Device Mode
WiFi Mode

WiFi SSID
WittelB

WiFi Password
augmatic@2020

Broker IP / Host
broker.emqx.io

Publish Topic
tcp100

Username
admin

MQTT Password
public

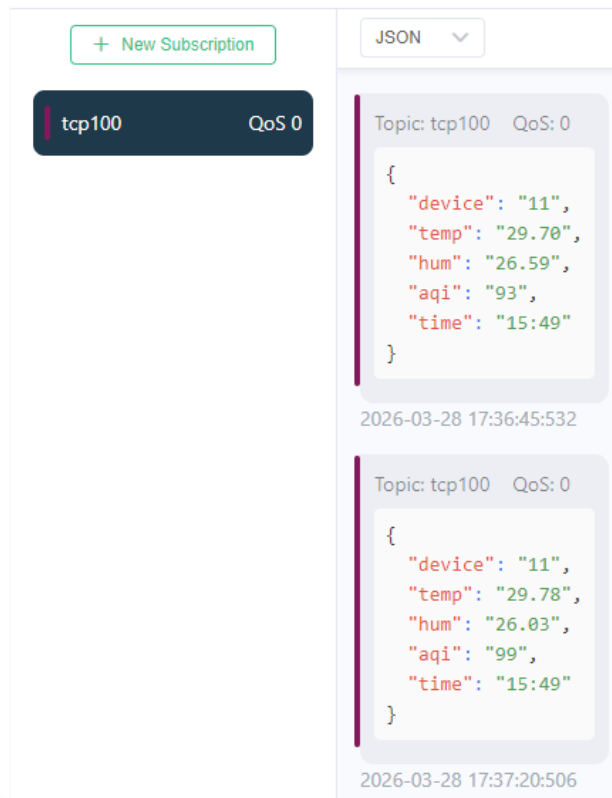
Device ID
123

Port
1883

Publish Interval
10 Second

Save Config

Image: 3



+ New Subscription

JSON

tcp100 QoS 0

Topic: tcp100 QoS: 0

```
{
  "device": "11",
  "temp": "29.70",
  "hum": "26.59",
  "aqi": "93",
  "time": "15:49"
}
```

2026-03-28 17:36:45:532

Topic: tcp100 QoS: 0

```
{
  "device": "11",
  "temp": "29.78",
  "hum": "26.03",
  "aqi": "99",
  "time": "15:49"
}
```

2026-03-28 17:37:20:506

Image: 4

- Enter the local Wi-Fi credentials as shown in above image3 so the device can connect to that Wi-Fi network.
- Enter the MQTT credentials to transmit the sensor data to the cloud.
- Image 4 shows the sensor data at cloud for the selected topic.

NOTE: If the device is selected in **WIFI Mode**, the web interface will be accessible at any time, but it will not connect to Modbus RTU.

Contact Information

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